

NHS Severe Patient Harm Forecasting

SPHERE-PPL Contest Submission Report

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Introduction

This report describes an XGBoost-based forecasting pipeline developed for the SPHERE-PPL Forecasting Contest. The objective is to forecast daily estimated avoidable deaths associated with Emergency Department admission delays across the Bristol NHS system.

Predictions use operational information available up to the midday cutoff on day D . Target-derived features are restricted to values reported at least three days earlier, thereby respecting the reporting-lag constraint and avoiding use of recently unavailable outcome information. The workflow is reproducible from the pre-computed outputs in `submission/development/` and `submission/validation/`.

Reproducibility note: The report reads only precomputed outputs from `submission/development/` and `submission/validation/`. Run `Rscript run_forecast_xgb.R development` and `Rscript run_forecast_xgb.R validation` before knitting.

Development Results

Data Preparation and Features

The development dataset was converted to a daily modelling matrix. Observations recorded by midday were assigned to the current day, whereas later observations were assigned to the following forecast day. Predictor missingness was handled using time-series imputation and fallback rules; the target was retained separately for model fitting and retrospective evaluation.

The feature set included operational indicators, calendar variables, seven-day rolling means, target lags of at least three days, and selected interaction terms. Hierarchical correlation-based screening reduced redundant operational predictors before model fitting.

Table 1: Prepared modelling dataset: development

n_days	n_predictors	start_date	end_date
916	360	2023-03-30	2025-09-30

Table 2: Number of predictors by transformation type: development

transformation	n_predictors
None (Raw Operational Indicators)	180
Rolling Windows Summaries	172
Target Lags (≥ 3 days)	4
High-Order Interactions	4

Backtesting and Retrospective Validation

Development performance was evaluated with rolling forecast origins and a 10-day prediction horizon. Separate XGBoost configurations were used for days 1–5 and days 6–10, while maintaining the target-lag restriction in every test window.

The development MSE was 0.0158 for days 1–5 and 0.0175 for days 6–10. Runtime was approximately 14.17 minutes for 151 forecast windows.

Table 3: Development MSE by forecast horizon

model	mse_1_5	mse_6_10
Hybrid XGBoost (development)	0.0158	0.0175

Table 4: Window-level MSE distribution: development

Horizon	Mean MSE	Median MSE
1-5 days (Short-term)	0.0158	0.0135
6-10 days (Long-term)	0.0175	0.0159

Validation Results

The final pipeline was rerun on the amended validation dataset. Observed estimated avoidable deaths were available from 1 October 2025 to 17 February 2026, enabling retrospective error calculation.

The validation evaluation included 131 rolling origins, from 30 September 2025 to 7 February 2026. Each origin generated a 10-day forecast, yielding 1,310 forecast–outcome pairs; all were available and finite.

Validation Data Summary

Table 5: Prepared modelling dataset: validation

n_days	n_predictors	start_date	end_date
1097	352	2023-03-30	2026-03-30

Table 6: Predictors by transformation type: Validation

Transformation Type	Number of Predictors
None (Raw Operational Indicators)	176
Rolling Windows Summaries	168
Target Lags (≥ 3 days)	4
High-Order Interactions	4

The mean validation MSE was 0.1258 for days 1–5, 0.1812 for days 6–10, and 0.1324 for the requested days 1–6 metric. The corresponding median MSE values were 0.0961, 0.1010, and 0.0961. The larger mean than median indicates that a small number of forecast windows had comparatively higher errors.

Table 7: Validation MSE by forecast horizon

model	mse_1_5	mse_6_10
Hybrid XGBoost (validation)	0.1258	0.1812

Table 8: Window-level MSE distribution: validation

Horizon	Mean MSE	Median MSE	Minimum MSE	Maximum MSE
1-5 days (Short-term)	0.1258	0.0961	0.0171	0.8566
6-10 days (Long-term)	0.1812	0.1010	0.0028	1.4319
1-6 days (Requested assessment metric)	0.1324	0.0961	0.0232	0.7914

Computational Performance

The development and amended validation executions used the same modelling framework and fixed hyper-parameters. Development comprised 151 forecast windows and validation comprised 131. The validation run required 13.84 minutes in total, which is within the one-hour computational constraint.

Table 9: Performance and computational summary

Dataset	MSE 1-5d	MSE 6-10d	Total Forecasts	Total Runtime (min)
Development	0.0158	0.0175	151	14.17
Amended validation	0.1258	0.1812	131	13.84

Submission Output

The submission includes `pred_matrix.csv`, containing the 10-day forecasts, and `mse_summary.csv`, containing one row for each of the 131 amended-validation forecast windows. The latter reports `forecast_id`, the forecast dates, `mse_1_5`, `mse_6_10`, and the requested `mse_1_6` metric.